

Behavioral Biometric Analysis for Task and Keystroke inference for task recognition

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Abstract—**L^AT_EX**.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Wearable devices such as smartwatches continuously collect motion-sensor data (e.g., accelerometer and gyroscope) to support health monitoring and activity-aware applications. The same sensing capability can also create privacy risk: prior work has shown that wrist-motion traces can leak typing-related information when applications can access sensor streams without strict controls [1], [2].

Several studies have already demonstrated this exact feasibility of motion-sensor keystroke inference attacks using machine learning techniques. One proposed defense mechanism is to block sensor access while keyboard activity is detected. This creates a serious usability problem however: disabling sensors interrupts legit applications that rely on motion data like health monitoring and fall detection. As a result of this, the user must choose between protecting their privacy and maintaining the core functionality of their wearable devices. This threat creates a practical defense challenge. Strong restrictions (e.g., disabling sensor access during sensitive activity) can reduce leakage, but may also disrupt legitimate smartwatch functionality. Differential privacy (DP) offers a principled way to perturb data, but in wearable settings the key question is empirical: how much attack degradation is achieved at a given perturbation level, and how much utility is lost for non-sensitive tasks.

This paper presents an empirical design-and-evaluation study of a fixed-noise DP defense implemented with the IBM Differential Privacy Library (`diffprivlib`) under a reproducible protocol. Our pipeline has three stages: (1) attack-feasibility modeling, (2) feature-level DP perturbation with fixed privacy budgets, and (3) privacy-utility evaluation.

Our claims are intentionally scoped. We evaluate attack components using two public datasets and do not claim a single end-to-end keystroke-content recovery pipeline. Specifically, WISDM is used for typing-likelihood task inference from smartwatch motion features, and Keystroke100 is used for closed-set *who-is-typing* identification from keystroke dynamics. We report measured privacy-utility tradeoffs under these defined conditions.

II. LITERATURE REVIEW

The privacy risks of unrestricted motion-sensor access on wearables are well established in the security literature. Early work showed that inertial signals from wrist-worn devices are not merely benign context signals; they can encode behavior linked to user input actions. This is especially important for smartwatches, where accelerometer and gyroscope data are often broadly available to applications, including those that do not need high-fidelity motion streams for their core function.

Wang *et al.* (MoLe) [1] investigated whether smartwatch motion sensors can leak information about laptop keyboard input. Their study combined motion-signal processing with language constraints and reported substantial leakage under their evaluated conditions (including left-wrist watch placement). Rather than framing the result only as per-key classification, MoLe also evaluated word-level narrowing, showing that typed-word candidates can be reduced meaningfully. The broader implication is that raw inertial streams can support side-channel inference even without direct keyboard or screen access. For our study, this paper is foundational because it demonstrates that smartwatch motion channels should be treated as potentially sensitive in typing-adjacent contexts.

Maiti *et al.* [2] extended this line of work and provided a more explicit smartwatch keystroke-inference attack framework in which wrist-motion characteristics are mapped to keyboard transition patterns and matched against candidate words. Their evaluations reported strong attack effectiveness in controlled settings, reinforcing that practical exploitation is feasible when sensor access is unaudited. Importantly, they also proposed a context-aware protection mechanism that can disable or downgrade motion-sensor access when typing is detected. This defense direction is highly relevant because it makes the privacy-utility tension explicit: strict blocking may reduce leakage but can interfere with legitimate sensing applications that depend on continuous motion data.

Differential privacy (DP) is a natural candidate for reducing inference risk while retaining utility, but wearable constraints complicate direct deployment. SeRaNDiP [3] addressed this issue in wearable *community sensing* by leveraging inherent sensor random noise rather than relying only on additional software-generated noise at the user device. Their work em-

phasizes that conventional DP pipelines can be expensive on constrained hardware and explores hardware-configuration effects (e.g., sensing settings) on usable noise characteristics. Reported results include lower overhead than baseline implementations and formal distributed-DP framing for community sensing. However, SeRaNDiP’s target setting is population-level community sensing rather than a keystroke side-channel attacker focused on an individual typing context. Thus, it is highly relevant methodologically, but not a direct substitute for a keystroke-risk defense evaluation.

Adaptive DP research in adjacent domains also informs this area. AGC-DP [4] applies adaptive Gaussian clipping in federated learning to improve privacy–utility behavior during model training. Its adaptation is driven by training dynamics in distributed optimization, not by smartwatch activity context or side-channel threat states. The paper is still useful to cite because it demonstrates that privacy parameters and clipping/noise schedules materially affect utility outcomes in practice. For our setting, the key takeaway is not to transfer the method directly, but to adopt rigorous privacy–utility evaluation discipline when selecting perturbation settings.

Zhao *et al.* (AdaTraj-DP) [5] proposed an adaptive DP framework for trajectory-data publishing that combines context-aware sensitivity modeling with hierarchical aggregation and dynamic budget allocation. Their setting is spatiotemporal trajectory release (e.g., location traces), and the reported benefits focus on preserving spatial consistency and query utility under varying privacy budgets. This differs from smartwatch typing-risk mitigation, where the primary concern is side-channel leakage from inertial features and downstream attacker performance. Nevertheless, AdaTraj-DP is relevant as evidence that context structure and budget design are central to practical privacy–utility tradeoffs.

Beyond DP mechanisms, Wang *et al.* (Hippo) [6] introduced a hierarchical information-dissociation approach for activity-sensing privacy, using a diffusion-based framework to control the disclosure of sensitive attributes and activity granularity. Hippo targets fine-grained controllable disclosure in mobile sensing and does not specifically optimize for keystroke side-channel robustness. Its contribution is conceptually adjacent: privacy control should be goal-aware, and not all sensitive information in sensing streams is of the same type. For our work, Hippo helps position the broader landscape in which side-channel mitigation is one specific privacy objective among several in sensing systems.

A. Gap Analysis and Our Contribution

Across these works, three points are clear. First, smartwatch motion sensors can leak typing-related information [1], [2]. Second, privacy protections that simply block sensing can create functional costs [2]. Third, DP-based protections are promising but often evaluated in different domains (community sensing, federated learning, trajectory release) with different adversary models and utility objectives [3], [4], [5], [6].

Our study is positioned as an empirical, implementation-grounded evaluation in this gap: we assess fixed-noise DP perturbation for smartwatch keystroke-risk proxies using IBM `diffprivlib`, under a reproducible protocol and explicit privacy–utility reporting. We do not claim a new DP theorem or a universal defense. Instead, we contribute a controlled benchmark-oriented analysis of how fixed privacy budgets affect (i) attack feasibility proxies and (ii) non-sensitive utility in this smartwatch-centered setting.

III. METHODOLOGY

Our methodology follows a three-stage pipeline: (1) attack demonstration through task inference and user identification, (2) defense using IBM differential privacy mechanisms, and (3) evaluation of the privacy–utility tradeoff.

We use two publicly available datasets. The first is the WISDM Smartphone and Smartwatch Activity and Biometrics Dataset [7]. We use smartwatch accelerometer and gyroscope ARFF feature files. In the released ARFF format, each modality file contains 93 columns (including metadata fields such as activity and class). In our implementation, watch accelerometer and watch gyroscope windows are merged by activity/window alignment, and metadata columns are removed for modeling, yielding 182 numeric input features per sample (91 accel + 91 gyro). We label F (typing) and Q (writing) as typing-related positives and all remaining activities as negatives for binary typing-likelihood detection.

The second dataset is Keystroke100 [8], which contains keystroke dynamics from 100 users with 10 samples per user. As in the original dataset design, all users type the same password, so this dataset is used for *who*-is-typing identification (not *what*-is-typed inference). In the main pipeline, we use 13 tabular features per sample: 7 latency features and 6 pressure-summary features (mean, standard deviation, maximum, minimum, peak, and duration).

A. Dataset Separation and Attack-Chain Scope

The WISDM and Keystroke100 datasets are independent and are used for complementary feasibility analyses rather than a single chained attack. WISDM supports typing-vs-non-typing task inference from smartwatch motion features, while Keystroke100 supports closed-set user identification from keystroke dynamics where all users type the same password. No cross-dataset identity linkage or cross-dataset transfer learning is performed; conclusions should therefore be interpreted as evidence of attack components, not as a full end-to-end keystroke-content inference pipeline.

B. Stage 1: Attack Demonstration

The first stage of our methodology establishes a baseline attack by demonstrating two capabilities: detecting when a user is typing from smartwatch sensor data, and identifying individual users from their keystroke dynamics.

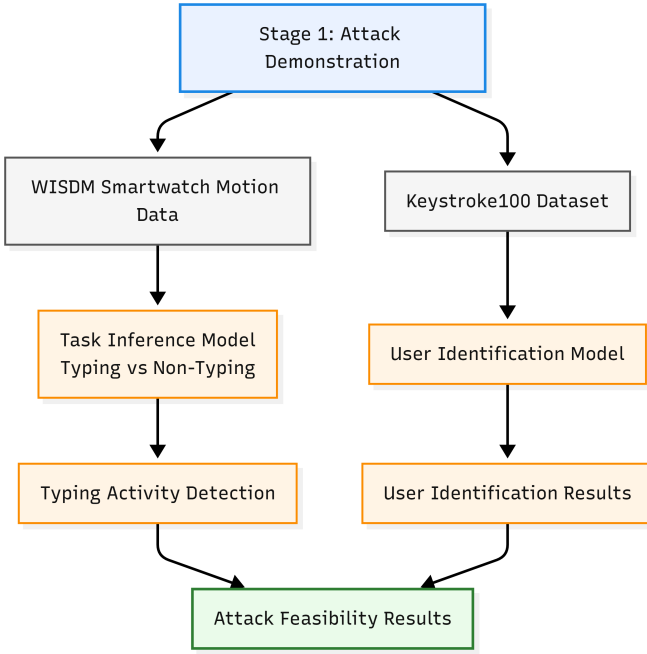


Fig. 1. Attack Demonstration flowchart

C. Datasets

1) *Task Inference*: For task inference, we use the WISDM smartwatch data to train a classifier that distinguishes between typing-related activities and non-typing activities. We formulate binary task inference on WISDM as typing-related vs. non-typing, where labels F and Q are positive and all remaining labels are negative. We evaluate Logistic Regression, Random Forest, and SVM with RBF kernel. Standardization is applied in the Logistic Regression and SVM pipelines. Data is split 70%/15%/15% (train/validation/test) using subject-disjoint splitting to test cross-user generalization. We report accuracy, precision, recall, F1-score, and ROC-AUC. The final task model is selected on validation performance (primary key: F1; tie-breaker: ROC-AUC), then refit on train+validation before test evaluation.

2) *User Identification*: For user identification, we use the Keystroke100 dataset to train a model that identifies which of the 100 users is typing based on their keystroke dynamics. We train multiclass user-identification models (100 classes) using the 13-feature representation described above. We evaluate k-Nearest Neighbors, Random Forest, and SVM with RBF kernel. Standardization is applied in the kNN and SVM pipelines. We use a stratified 70%/15%/15% split (train/validation/test) by user label. We report top-1 accuracy, top-5 accuracy, weighted precision, weighted recall, weighted F1-score, and one-vs-rest mean Equal Error Rate (EER). The final model is selected on validation top-1 accuracy (tie-breaker: top-5 accuracy), then refit on train+validation before test evaluation.

These two models demonstrate attack feasibility in two parts: (1) detecting typing-related behavior from smartwatch motion features and (2) identifying users from keystroke

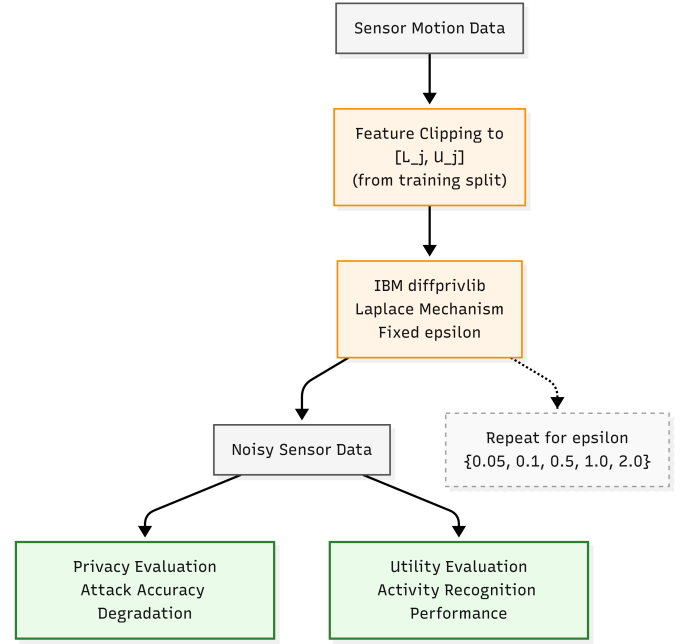


Fig. 2. Differential privacy and Evaluation flowchart

dynamics. As the datasets are separate, this is a feasibility study rather than a single end-to-end attack chain.

D. Stage 2: IBM Differential Privacy Defense

1) *Differential Privacy Setup*: We implement feature-level perturbation using the IBM Differential Privacy Library (diffprivlib) with the Laplace mechanism. For each numeric feature j , values are clipped to fixed bounds $[L_j, U_j]$ computed from the training split only and reused unchanged for validation and test data.

The per-feature Laplace scale is:

$$b_j = (U_j - L_j)/\epsilon.$$

2) *Fixed Privacy-Budget Policy*: Each run uses one fixed privacy budget ϵ for all windows and all activities. No activity-dependent switching is applied, and Stage 1 outputs are not used to modify ϵ during inference.

3) *Baselines*: We compare against: (1) a clean (no-defense) baseline, and (2) IBM fixed-noise DP settings at different ϵ values.

E. Stage 3: Evaluation

1) *Privacy Evaluation*: We evaluate two attacker settings: (1) clean-trained attacker tested on noisy test data, and (2) attacker-aware retraining (attacker trained on noisy train+validation and tested on noisy test). We report attack accuracy degradation relative to clean-data baselines.

2) *Utility Evaluation*: Utility is measured on non-typing activity recognition. A Random Forest utility model is trained on clean non-typing train+validation data, then evaluated on clean test data and IBM-noisy test data. Utility loss is measured as clean utility accuracy minus noisy utility accuracy.

3) *Privacy-Utility Tradeoff Sweep*: We run a fixed- ε parameter sweep:

$$\varepsilon \in \{0.05, 0.1, 0.5, 1.0, 2.0\}.$$

For each setting, we report task-attack accuracy on noisy data, privacy gain (clean accuracy minus noisy accuracy), non-typing utility accuracy, and utility loss.

F. Implementation

The pipeline is implemented in Python using `numpy`, `pandas`, `scipy`, and `scikit-learn`. Differential privacy perturbation is implemented with IBM `diffprivlib` Laplace mechanisms. Experiments use fixed random seeds for reproducibility.

IV. RESULTS

V. CONCLUSION

A. Future Work

ACKNOWLEDGMENT

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